

Planning the Next Evidence Acquisition Step: An Offline Framework for Active Scientific Discovery

Scientific Evidence Planner Project

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Abstract

Most work on machine-assisted science focuses on generating scientific questions or executing experiments autonomously. We address the step in between: given an existing scientific question, a set of competing hypotheses, and the evidence currently available, *what should we investigate next to maximize expected scientific progress?* We present an offline, domain-independent planning framework that represents evidence as a typed, confidence-weighted graph; identifies hypotheses with missing, weak, or non-independent evidence; generates generic candidate evidence-acquisition actions; and ranks them by a configurable utility combining expected information gain, novelty, tractability, and cost. Beliefs over hypotheses are revised with a modular Bayesian updater, closing the plan–observe–update loop without ever executing actions. Because the planner is offline, it can be evaluated honestly against history: a benchmark truncates the evidence graph at a time cutoff, hides later evidence as ground truth, and asks whether the recommended actions would have resolved the question earlier. On a worked case study from exoplanet atmospheric characterization, the planner recommends the independent follow-up observation and retrieval re-analysis that historically resolved the question, achieving a resolution efficiency of 1.0 and an expected entropy reduction of 0.32 bits over the hypothesis set. The framework is released as a reference implementation with seven evaluation metrics, full configurability, and no domain assumptions in code.

1 Introduction

Scientific progress is rarely limited by the ability to ask questions; it is limited by the cost of acquiring the evidence that discriminates between competing answers. A researcher facing an open question — is a reported effect real, an artifact of the analysis, a confounder, or an instrument systematic? — must decide where to spend the next unit of effort: re-read the literature, re-run an analysis with different assumptions, compare independent datasets, run a simulation, or invest in a costly new observation. This decision is made under uncertainty, under resource constraints, and usually informally.

This paper treats that decision as a planning problem. We assume the scientific question already exists (produced by a human or by an upstream question-generation system) and ask only: *given the current evidence, which evidence-acquisition step maximizes expected scientific progress?*

Our design commitments differ from much recent work on autonomous science:

- **Offline.** The system is a planner, not an agent. It does not browse, does not call instrument APIs, and does not execute experiments. Everything works from static datasets, which makes the system auditable and makes rigorous historical evaluation possible.

- **Domain-independent.** No scientific field is hardcoded. Domain knowledge enters only through data (questions, hypotheses, evidence graphs) and configuration (action catalogs, weights).
- **Modular.** Every component — hypothesis generation, gap analysis, action generation, each utility term, belief updating, evaluation — sits behind a small interface and can be replaced independently.

Contributions. (1) A formalization of next-step evidence planning over typed, confidence-weighted evidence graphs with provenance; (2) a gap analysis that flags hypotheses with missing, sparse, weak, or non-independent evidence; (3) a configurable utility decomposition — expected information gain via preposterior analysis, provenance-based novelty, resource-aware tractability, and cost; (4) a historical-simulation benchmark with seven metrics that evaluates whether a planner would have resolved a question earlier; and (5) an open reference implementation with unit tests and a worked case study.

2 Problem Statement

Let q be a scientific question and $\mathcal{H} = \{h_1, \dots, h_n\}$ a set of mutually exclusive, exhaustive competing hypotheses with probabilities $p = (p_1, \dots, p_n)$, $\sum_i p_i = 1$. Let $G = (V, E)$ be the current *evidence graph* (Section 3.1). Let $\mathcal{A}$ be a set of candidate evidence-acquisition actions, each with a type, relative cost, relative time, required resources, and a target hypothesis.

The planner must return a ranking of $\mathcal{A}$ (top- K recommendations) maximizing a utility

$$U(a) = w_{\text{ig}} \text{IG}(a) + w_{\text{nov}} \text{Nov}(a) + w_{\text{tr}} \text{Tr}(a) - w_c C(a), \quad (1)$$

where IG is the expected reduction in hypothesis uncertainty, Nov measures independence from existing evidence, Tr measures feasibility, C is cost, and the weights w are configurable. After an action is (externally) executed and its outcome observed, hypothesis probabilities are revised and the cycle repeats. The framework plans and updates; it never executes.

3 Framework

3.1 Evidence Graphs and Provenance

Evidence is a directed multigraph with five node types — *paper*, *observation*, *dataset*, *claim*, *hypothesis* — and four edge types — *supports*, *contradicts*, *derived_from*, *observes* — each edge carrying a confidence in $[0, 1]$. Claims point at the hypotheses they support or contradict; derived nodes point at their origins (claim $\rightarrow$ paper $\rightarrow$ dataset); observations point at what they observe. Nodes optionally carry timestamps, which enables historical simulation (Section 4).

Two queries are load-bearing. First, $\text{evidence}(h)$ returns the supports/contradicts edges incident to a hypothesis. Second, $\text{roots}(v)$ follows *derived_from* edges transitively to the root sources of a node. Two claims that descend from the same dataset are *not* independent evidence, no matter how many papers separate them; this notion of provenance drives both gap analysis and novelty.

3.2 Hypothesis Generation

Hypotheses are structured objects (identifier, description, prior, supporting and contradicting evidence references). The reference implementation provides two interchangeable generators behind

one interface: a *static* generator that loads curated hypothesis sets keyed by question (the normal path for imported datasets), and a *template* generator producing four generic methodological alternatives applicable to any empirical claim: the effect is real; it is an artifact of the analysis method; it is confounded by an unmodeled factor; it is an instrument or measurement systematic. Priors are normalized, defaulting to uniform.

3.3 Evidence Gap Analysis

For each hypothesis the analyzer emits the first applicable gap:

1. **No evidence.** No edge touches the hypothesis (gap confidence 1.0).
2. **Sparse evidence.** Fewer than $k_{\min}$ evidence edges; confidence $0.5 + 0.5(1 - \text{count}/k_{\min})$.
3. **Weak support.** Summed support confidence below a threshold $s_{\min}$; confidence $0.4 + 0.4\delta$ where δ is the relative deficit.
4. **No independent corroboration.** All evidence descends from a single provenance root (confidence 0.78): formally, $|\bigcup_{v \in \text{sources}(h)} \text{roots}(v)| \leq 1$.

The output is a gap report: hypothesis, missing-evidence description, confidence, and a human-readable reason, sorted most-confident first. All thresholds are configurable.

3.4 Candidate Action Generation

Actions are instantiated from a generic, YAML-configurable catalog of seven types: read paper, search archive, re-run analysis, collect observation, run simulation, compare datasets, search review. Each template carries a relative cost and time in $[0, 1]$ and required resource tags. For each gap, the generator instantiates the subset of types that plausibly closes that kind of gap — independence gaps receive independent-source actions (collect observation, compare datasets, search archive); weak-support gaps receive re-analysis actions; evidence-free hypotheses receive broad search actions — and records the gap’s reason and confidence in the action’s metadata. Nothing in the catalog or the mapping references any scientific domain.

3.5 Utility Estimation

Each term of Eq. (1) is an independent estimator behind a common interface, injected into the combiner; any term can be replaced without touching the others.

Expected information gain (preposterior analysis). An action a targeting hypothesis h_t is modeled as producing a binary outcome $o \in \{\text{favors } h_t, \text{ disfavors } h_t\}$ with a likelihood ratio r_a determined by the action type’s *discriminative power* (a configurable table; an independent observation separates hypotheses more sharply than reading a paper):

$$P(o^+ | h_t) = \frac{r_a}{r_a + 1}, \quad P(o^+ | h_j) = \frac{1}{r_a + 1} \quad (j \neq t). \quad (2)$$

The estimate is the expected entropy reduction

$$\text{IG}(a) = H(p) - \mathbb{E}_o[H(p | o)] \geq 0, \quad (3)$$

with H the Shannon entropy in bits and the posterior computed by Bayes’ rule for each outcome. This is exact for the two-outcome model and degrades gracefully: actions with likelihood ratio 1 yield zero gain, as do actions with no target among the live hypotheses.

Novelty (provenance independence). Each action type has a base source novelty $\nu_a \in [0, 1]$ (new observation = 1.0; re-reading literature ≈ 0.3). The base is boosted by the provenance concentration of the target’s existing evidence:

$$\text{Nov}(a) = \min\left(\nu_a\left(1 + \frac{1}{\max(|R_t|, 1)}\right), 1\right), \quad (4)$$

where R_t is the set of provenance roots behind the target hypothesis’s evidence. If everything known descends from one dataset, independent evidence is maximally valuable.

Tractability. $\text{Tr}(a) = (1 - \text{time}_a) \cdot \rho^m$, where m is the number of required resources not declared available and ρ (default 0.5) is the missing-resource penalty. With no declared resource set, all resources are assumed available.

Cost. The action’s configured relative cost, passed through.

Calibration. Information gain is measured in bits and is typically well below 1 for a single action, while the other terms are $[0, 1]$ scores; the default configuration therefore sets $w_{\text{ig}} = 10$ to place the terms on a comparable scale, with all weights exposed in YAML.

3.6 Planning

The planner is deliberately thin: it scores every candidate with the injected utility estimator, sorts by total utility, truncates to top- K , and attaches to each recommendation its utility breakdown, its target hypothesis, and a reasoning string naming the dominant term. All ranking policy lives in the (replaceable) utility estimator.

3.7 Belief Updating

New evidence arrives as a likelihood vector $L = (P(e | h_1), \dots, P(e | h_n))$; hypotheses absent from the vector receive a neutral likelihood of 1. The default updater is exact discrete Bayes,

$$p'_i = \frac{p_i P(e | h_i)}{\sum_j p_j P(e | h_j)}, \quad (5)$$

with two safeguards: evidence impossible under every hypothesis leaves the distribution unchanged (it is uninformative for this hypothesis set), and the update is non-destructive, recording the evidence identifier on each hypothesis’s supporting or contradicting list according to the direction of the probability change. The updater is a protocol; richer models (soft evidence, correlated hypotheses, hierarchical priors) can be substituted without touching callers.

4 Offline Evaluation by Historical Simulation

An offline planner can be evaluated the only way that avoids self-grading: against history. The benchmark takes a *full* timestamped evidence graph and a cutoff τ :

1. Filter the graph to nodes with timestamp $< \tau$ (untimestamped structural nodes are kept; edges touching hidden nodes are dropped). The planner sees only this sub-graph.
2. Run any planner pipeline (a two-method protocol) on the visible sub-graph, producing recommendations and the hypothesis set used.

3. Express post-cutoff evidence as ground-truth records: which hypothesis it bore on, which action type historically produced it, and its per-hypothesis likelihoods.
4. A ground-truth item is *reached* if some recommendation matches its action type and target hypothesis; reached items are replayed through the belief updater to measure the uncertainty the plan could have removed.

The central question is: *did the planner recommend the actions that would have resolved the question earlier?*

4.1 Metrics

Top- K utility. Mean total utility of the top- K recommendations.

Hypothesis entropy reduction. $H(p_{\text{before}}) - H(p_{\text{after}})$ after replaying reached ground truth — what history says the plan was worth.

Average information gain. Mean *expected* gain across recommendations — what the planner believed; comparing it with realized entropy reduction exposes over- or under-confidence.

Action diversity. Distinct action types over number of recommendations, penalizing degenerate plans.

Coverage of competing hypotheses. Fraction of hypotheses targeted by at least one recommendation; a good plan probes the space of explanations, not just the front-runner.

Resolution efficiency. Fraction of ground-truth evidence reached by the plan; 1.0 means every piece of evidence that historically resolved the question was on the recommended path.

Expected utility. Rank-discounted utility with weights $1/\text{rank}$ (normalized), modeling that earlier recommendations are likelier to be executed.

5 Case Study

We instantiate the framework on a question from exoplanet atmospheric characterization — “*Is the reported low water abundance real or a retrieval artifact?*” — stressing that the domain lives entirely in the data files; the code is untouched. Four hypotheses compete: the low abundance is real ($p = 0.3$); it is a retrieval artifact ($p = 0.3$); it is a cloud-composition degeneracy ($p = 0.2$); it is an instrument systematic ($p = 0.2$). The pre-cutoff graph (cutoff $\tau = 2020$; the planner sees 10 of 13 nodes) contains a discovery paper and an independent re-analysis, both derived from the *same* transmission spectrum dataset — textbook provenance concentration. The hidden ground truth is the mid-infrared follow-up observation (2023) that independently confirmed the low abundance.

Gap analysis flags the “real” hypothesis for lacking independent corroboration (all evidence descends from one dataset), the “artifact” hypothesis for sparse evidence, and the remaining two for having no evidence. The planner’s top recommendations are re-running the retrieval under alternative settings (targeting the artifact hypothesis) and collecting an independent observation (targeting both leading hypotheses) — precisely the two moves the community historically made.

Table 1 summarizes the run. Resolution efficiency of 1.0 indicates the historically decisive evidence was on the recommended path; replaying it yields an entropy reduction of 0.32 bits over the four-hypothesis set. Coverage of 0.5 reflects that the two low-prior hypotheses did not make the

Table 1: Benchmark results for the case study (top- $K=5$, cutoff 2020, default configuration).

Metric	Value
Top- K utility	2.596
Hypothesis entropy reduction (bits)	0.319
Average information gain (bits)	0.173
Action diversity	0.600
Coverage of competing hypotheses	0.500
Resolution efficiency	1.000
Expected utility	2.616

top-5 under default weights — raising the coverage-relevant weights or K trades focus for breadth, and the benchmark makes that trade-off measurable.

The utility weights also behave as designed: with all weights at 1, cheap archive searches dominate the ranking and the decisive (expensive) observation is never recommended; the calibrated default ($w_{\text{ig}} = 10$) restores the balance between bits and $[0, 1]$ scores. We report this deliberately — the failure mode is exactly the kind of miscalibration the benchmark exists to expose.

6 Implementation

The reference implementation is pure Python (standard library plus PyYAML), organized as ten modules that communicate only through shared dataclasses and the graph interface; estimators and generators are injected, never imported across modules. All tunable behavior — utility weights, gap thresholds, the action catalog, discriminative powers, source novelties, resource sets — lives in YAML or documented constructor parameters. The test suite (48 unit tests) covers question loading, the graph, gap analysis, action generation, every utility component, planner ranking, belief updating, each evaluation metric, and the end-to-end benchmark. An experiment script reproduces Section 5 in one command.

7 Limitations and Future Work

- **Two-outcome information gain.** The preposterior model collapses an action’s result to favors/disfavors with a per-type likelihood ratio. Multi-outcome models, and ratios conditioned on the specific hypothesis pair, are natural extensions behind the same interface.
- **Type-and-target ground-truth matching.** Resolution efficiency currently matches recommendations to hidden evidence by action type and target hypothesis; semantic matching would credit plans that reach the same evidence by a different route.
- **Exhaustive, exclusive hypothesis sets.** The Bayesian updater assumes the hypothesis set partitions the possibility space; an explicit “none of the above” mass is a small but valuable addition.
- **Static candidate space.** Actions are instantiated from gaps in the current graph; sequential planning over multi-step campaigns (with evidence-dependent branching) is the obvious next step, and the offline design makes it benchmarkable the same way.

- **Single case study.** The framework ships with one worked historical example; populating the benchmark with question sets and evidence graphs imported from upstream projects is ongoing work.

8 Conclusion

We presented an offline, domain-independent framework that turns “what should we investigate next?” into a concrete, configurable, and — crucially — historically evaluable planning problem. By separating evidence representation, gap analysis, action generation, utility estimation, and belief updating behind small interfaces, the framework serves as a reference implementation that individual research programs can adapt by swapping components and editing configuration, not code. The historical benchmark reframes evaluation from plausibility to counterfactual usefulness: a good evidence planner is one that would have gotten science to the answer sooner.